

Jacob Dentes

Cambridge, Massachusetts | jdentes10@gmail.com | github.com/Jacob-Dentes

Education

- Cornell University, College of Engineering**, BS in Computer Science Aug 2021 – Dec 2024
- GPA: 4.17/4.3, Summa Cum Laude, Dean's List (all semesters)
 - **Coursework:** Analysis of Algorithms (PhD-level), Optimization, Probability and Statistics, Data Structures and Functional Programming, Computational Complexity
- MIT, Operations Research Center**, PhD Student Aug 2025 – Present

Awards

- Merrill Presidential Scholar
- NSF Graduate Research Fellowship Program Honorable Mention

Research Experience

- Undergraduate Researcher** - Scheduling June 2023 – Aug 2023
- Reduced elective conflicts for the Mechanical Engineering department by 60%
 - Designed and implemented a complex mixed integer program (MIP), creating a tool for easy parameter modification to suit user needs
- Undergraduate Researcher** - Network Planning Jan 2024 – Nov 2024
- Developed a prediction model to estimate the probability of reception for radio transmissions, utilizing real-world data to enhance model accuracy
 - Implemented a cost-sharing model that ensured fair pricing for users based on their coverage needs, achieving a cost recovery ratio approximately twice as high as competing methods
 - Created a public digital toolkit for planning wireless networks
- Undergraduate Researcher** - Congestion Pricing Aug 2024 – Aug 2025
- Applying advanced optimization techniques to develop a congestion pricing model for Manhattan's Central Business District, focusing on reducing traffic congestion while ensuring equity across socioeconomic groups
 - Leveraged sparsity to enhance computational efficiency, aiming to make the model tractable for applications in New York City by reducing hot-loop operations from $O(n^2)$ to $O(n)$ for a 100 times speed-up
- Research Assistant** - Joint Replenishment Problem Aug 2025 – Present
- Applying first-order methods to solve large-scale inventory problems on GPUs
 - Implemented a model that competes with state-of-the-art general purpose solvers on large instances

Publications

- Bounding the Price-of-Fair-Sharing using Knapsack-Cover Constraints to guide Near-Optimal Cost-Recovery Algorithms** Sept 2024
- Sander Aarts, *Jacob Dentes*, David Shmoys, Manxi Wu
10.48550/arXiv.2309.16914
- A Data-Driven Model for LoRaWAN Connection Quality and Coverage** Nov 2024
- IEEE World Forum on Internet of Things (WF-IoT), 2024
0.1109/WF-IoT62078.2024.10811435

Public Code Projects

- Wireless Network Planner** GitHub
- Developed a suite of tools for planning an internet of things (IoT) network
 - Provides tools for training or fine-tuning a machine learning connectivity prediction algorithm
 - Formulates and solves large-scale MIPs to place wireless receivers
 - Supports fine-grained control for technical users and high-level scripts for non-technical users
 - Tools used: Python, Pytorch, Geopandas, Rasterio, SCIP, Gurobi, Sphinx

Sorting Algorithm Visualizer

GitHub

- Developed a cross-platform interface for visualizing six sorting algorithms using Python and Tkinter
- Tools Used: Python, Tkinter

N-Body Simulator

GitHub

- Worked on a team of four to build an N-body simulation sandbox with OCaml
- Created a linear algebra library for computing gravitational interactions
- Created a command line interface for defining simulation scenes
- Tools Used: OCaml

MTD(f)

GitHub

- Created a Rust library for solving two-player zero-sum games based on the MTD(f) algorithm
- Implemented a parallel ABDADA solver to solve subproblems
- Tools Used: Rust

Teaching Experience

AEW Facilitator

Jan 2022 – Dec 2022

- Taught calculus and object-oriented programming to weekly classes of about thirty students
- Wrote weekly problem sets and slides
- Practiced evidence-based group-learning strategies

Teaching Assistant - Engineering Applications of Operations Research

Jan 2023 - Dec 2025

- Hosted weekly office hours and lab sections
- Proctored exams, wrote problem set solutions, and graded course materials
- Refactored lab materials and programmed visualizations for in-class demos

Teaching Assistant Training

Sept 2024

- Attended a workshop on best teaching practices
- Learned research-based strategies for grading, communication, and conflict resolution

Technologies

Languages: Python, Rust, OCaml, C, Java, C#, Swift, SQL, JavaScript / CSS, LaTeX

Technologies: .NET, Numpy, Pandas, Geopandas, Scipy, Gurobi, SCIP, Pytorch, Tensorflow, JAX, PyO3, Jupyter Notebook, React, Tkinter, Unity, Bevy